python Day-19

Bitwise operators:-

-> Binary number -> we can able to perform few operation -> by using bitwise operators

① NOT ② And ③ or ④ ex or

Byte ①

-> memory efficient +

Bit ->

Binary ->

8 4 2 1

0 1 0 1

8Bit

And

8 4 2 1 -> ①

0

5 -> 0 1 0 1

3 -> 0 0 1 1

0 0 0 1 -> ① —

or

8 4 2 1

0 1 0 ~~0~~

0 0 1 1

0 ① ① ① -> ⑦

4 + 2 + 1 = ⑦

or

ex or:-

8 4 2 1

0 1 0 1

0 0 1 1

0 ① ① 0 ->

4 + 2 6

1 1 -> 0

0 0 -> 0

0 1 — 1

1 0 — ①

NOT

1 byte → 8 bits →

128	64	32	16	8	4	2	1
0	0	0	0	0	1	0	1
1	1	1	1	1	0	1	0

5

Sign (−) (+) NOT

1 → negative (−)

0 → positive

0000 0101

0 1 1 → 2's comp

0000 0110 → (−) 6

4 + 2

$-(-12+1)$

$-(-12+1)$

$+12-1 \rightarrow 11$

$\sim x = -(x+1)$

$\sim 12 = -(12+1)$

$= -13$

$\sim 5 = -(5+1)$

$= -6$

Bod

$\sim -12 = -(-12+1)$

$= 11$

Left shift (<<)

→ move bits towards the Left

$5 << 1 \rightarrow 1$ → move bits two Left with 1 time

$n << n$

$= n \times 2^n$

$n << n$

$n \times 2^n$

$5 \times 2 \rightarrow 10$

8 4 2 1

0 1 0 1

0 1 0 1 0

5×2^1

$8 + 2 \rightarrow 10$

right shift (>>)

$x = x/2^n$

$5>>1 = 5/2$ $\boxed{2\times2}$

$= \cancel{50}/2 = \cancel{>}\boxed{2}$

5>> 8 4 2 1

5 -> 0 1 0 1

0 0 1 0 ~~1~~

$\boxed{2}$ ✓

identity opratrn:-

~~is~~ , is not -> memory id check

-> it is used to check the

a = 10 -> 200 is

b = 10 -> 200

print(a is b) # True

print (a is not b) # false

200 is 200

200 is not 200

mutable -

immutable -

a = [10,20,30] } false

b = [10,20,30] -> false

memeber ship->

-> it is used to check the items present in the seq

a = [10,20;30]

print (10 in a) #

print (10 not in a) #

a = [10, 20, 30] refer

a = b

~~a = 20~~

print (a is b)

1r

c

a b ->

a = [10, 20, 30]

b = [10, 20, 30]

b0 = 20

pen

40

a = [10, 20, 30)

→ (in) → True
→ not in → (True)

print(10 in a) True
print(400 not in a) ~~False~~ ~~True~~

(a = b)

10
a = [10, 20, 30] ↗
10
b = [10, 20, 30) →
5 → chose →
print(a in not b)

(False)

Terinary operator (condition operator):-

' ' if condition True else ' '

Trenary e

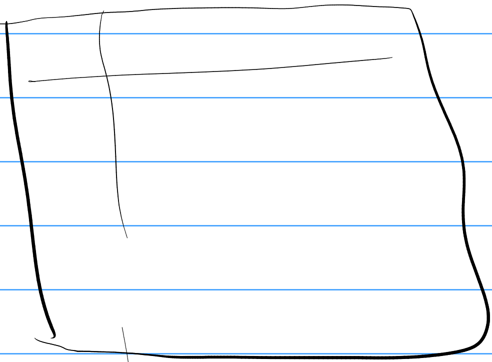
n = 6 (-1),
x = ('positive') if n > 1 (else) ('negative') →

print(x) in 6 > 1

posit   if (True)

→ operator precidnece → higen which operator exetry first → opestor pericide →

→ ( )
** +
- - -
* / % *
way



→ ( ) → prec.

a = int(10)

→ casting → T

type casting

└→ Datatype → another data type

Impricit casting

explisit casting

(+) byte

1+[1.

1+

Implisit (automic)

a = 1+2.5 → 3.5 → Error

print(a) = 3.5 →

↓

type error

explicit casting

a = 10 → float

print(float(a)) = 10.0